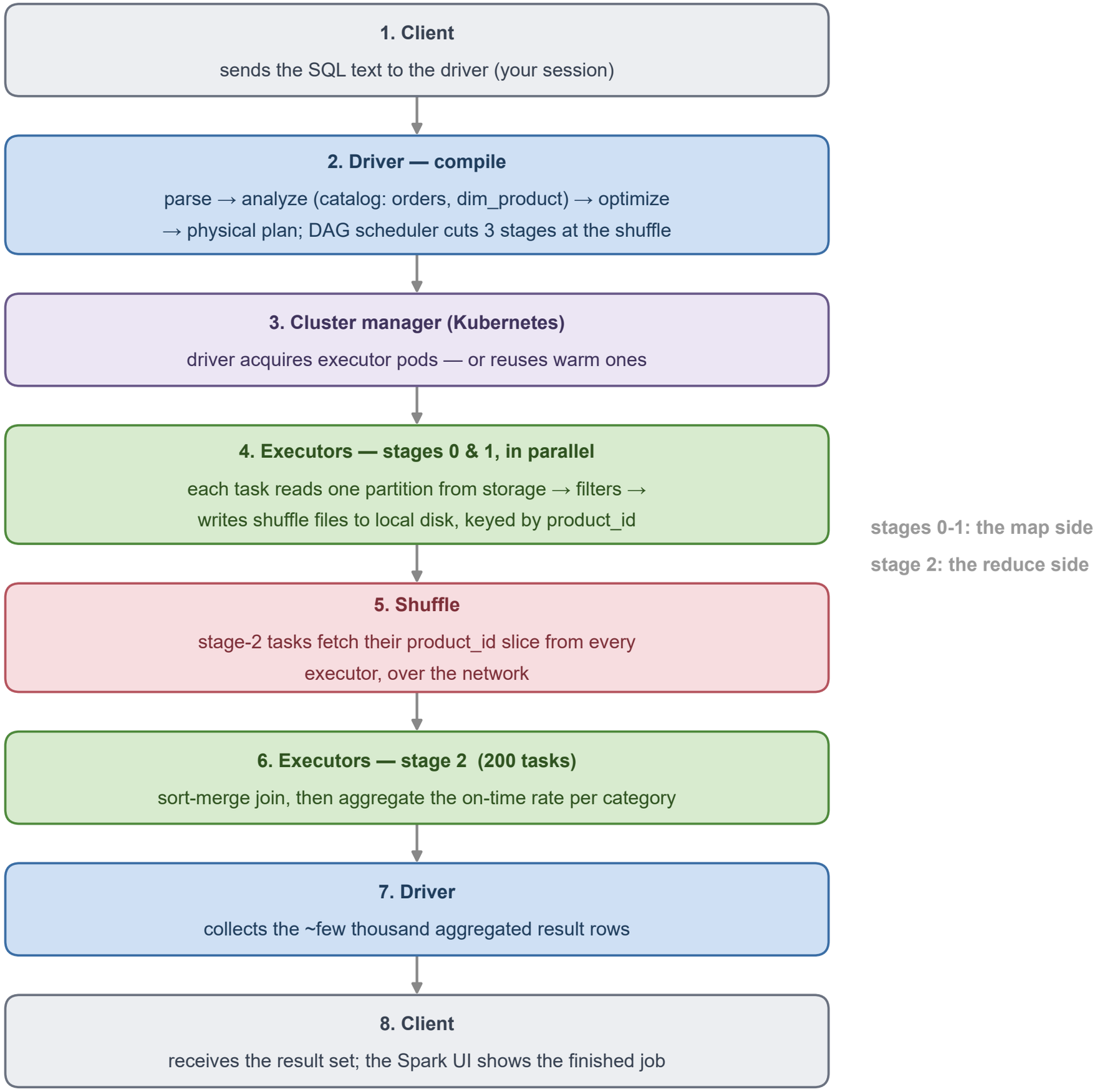

1. Client
sends the SQL text to the driver (your session)
2. Driver — compile
parse → analyze (catalog: orders, dim_product) → optimize
→ physical plan; DAG scheduler cuts 3 stages at the shuffle
3. Cluster manager (Kubernetes)
driver acquires executor pods — or reuses warm ones
4. Executors — stages 0 & 1, in parallel
each task reads one partition from storage → filters →
writes shuffle files to local disk, keyed by product_id
5. Shuffle
stage-2 tasks fetch their product_id slice from every
executor, over the network
6. Executors — stage 2 (200 tasks)
sort-merge join, then aggregate the on-time rate per category
7. Driver
collects the ~few thousand aggregated result rows
8. Client
receives the result set; the Spark UI shows the finished job
stages 0-1: the map side
stage 2: the reduce side